FaultLine

Detecting the Fracture Point — 14 Days Before It Happens

PlayHack ML Track - IIT Guwahati
Sports Board × Technical Board

Ctrl Freaks
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The Challenge

Task A (Binary Classification)

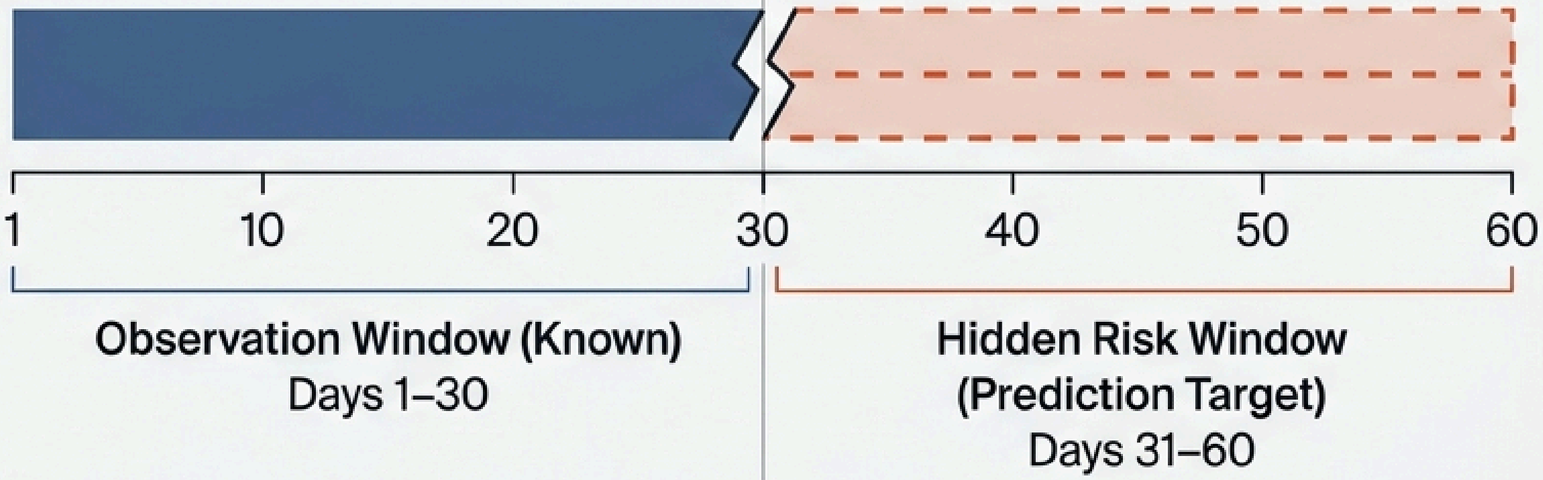
Predict whether an injury onset occurs in the hidden risk window (days 31–60).

Task B (Regression)

If injured, predict the onset day offset (1–30) and recovery duration (days).

Dataset Context

3,000 training athletes, 1,100 test athletes across 7 disparate data sources.

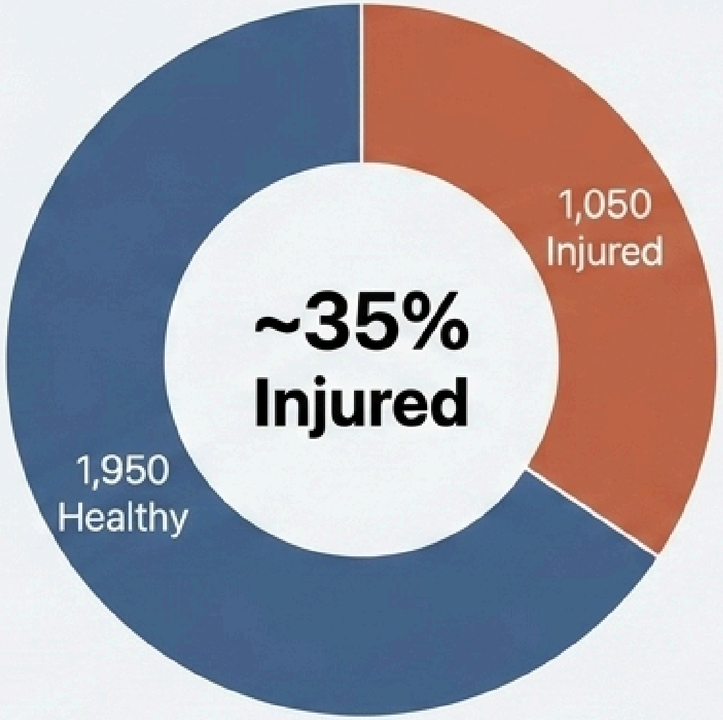


Exploratory Data Analysis

Data Landscape

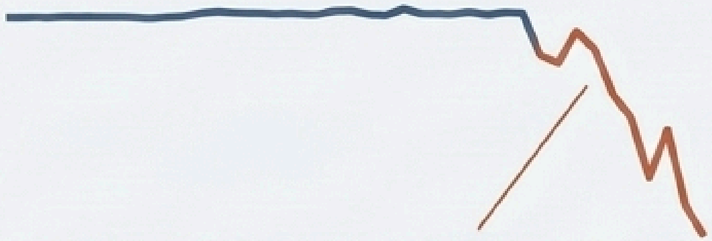
7 distinct sources: daily activity, sleep, heart rate, hourly calories/intensity/steps, training sessions, weight log.

Class Imbalance



Integrity & Signal

- Exceptional data quality with only 0.41% missingness (concentrated in weight/body-fat fields).



Crucial Observation: Wearable signals show clear degradation patterns in the final 7–14 days before injury onset.

Preprocessing & Data Integrity

Zero-Leakage Guarantee

✓	**Strict Window Enforcement**	Only days 1–30 used. Half-open intervals ensure all 24 hours of Day 30 are captured without leaking Day 31+.
✓	**Inclusive End-Date Fix**	Corrected a subtle underlying bug where inclusive end-dates silently dropped hours 01–23 on Day 30 for hourly sources.
✓	**Native NaN Handling**	No global imputation used. CatBoost handles NaNs natively, eliminating imputer leakage across cross-validation folds.
✓	**Categorical Integrity**	Sport, gender, dominant side, position, and team passed **directly** as native CatBoost categorical features.
✓	**System Checks**	`FEATURE_TABLE_VALIDATION = PASS` | `SOURCE_WINDOW_VALIDATION = PASS`

347 Engineered Features

10 Athlete Metadata
age, height, baseline weight, years playing, prior injuries

107 Daily Activity
steps, distance, active-minute bands, load ratios

42 Sleep
duration, efficiency, awake-in-bed, sleep debt

42 Heart Rate
daily avg, resting proxy (5th %ile), night HR

31 Hourly Calories
coverage-adjusted daily load, circadian share

31 Hourly Intensity
analogous structural load to calories

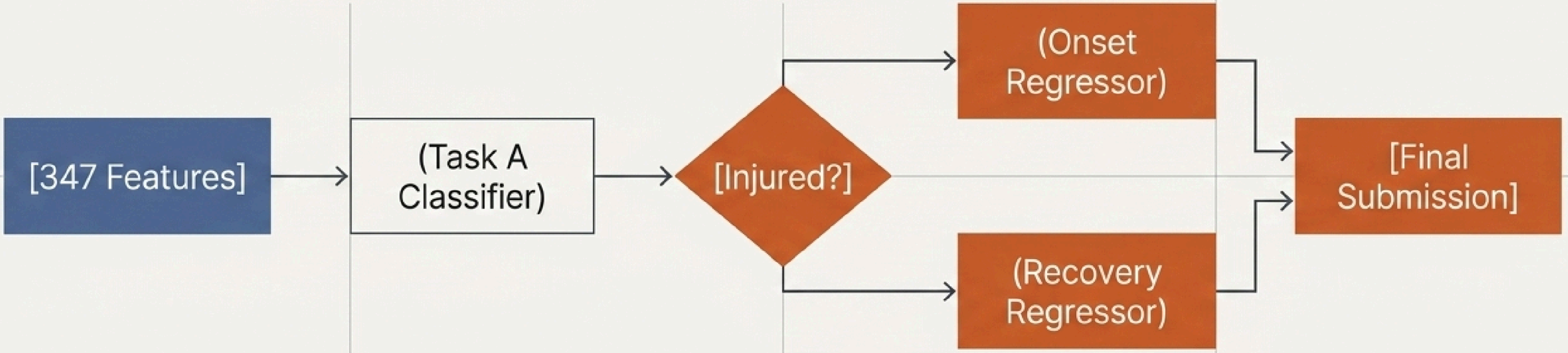
31 Hourly Steps
analogous structural load to calories

45 Training Sessions
load, streaks, monotony, strain, session mix

8 Weight Log & Interactions
latest BMI, load-vs-sleep, active-vs-sleep-debt

The **Athlete-Relative Philosophy:** Each athlete is compared against their own earlier behavior, never population statistics. Disjoint comparison windows prevent information overlap.

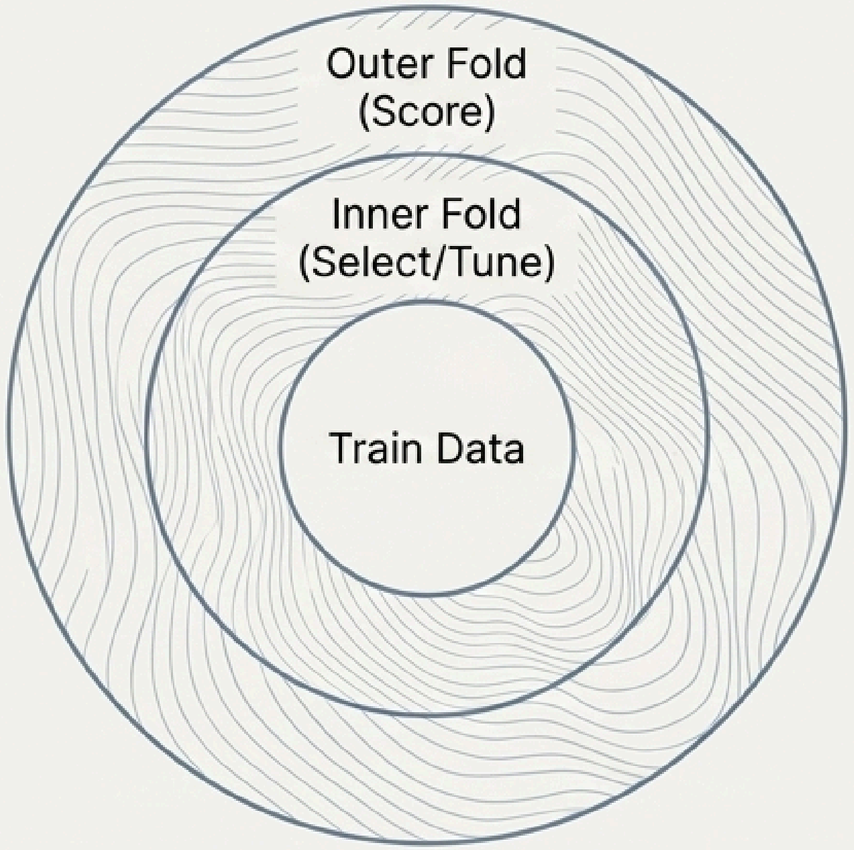
Three-Model CatBoost Pipeline



Model Type	Loss	Loss	Trained On	Trees	Depth	LR
Task A Classifier	CatBoost	Logloss (balanced)	3,000 athletes	50	6	0.03
Onset Regressor	CatBoost	MAE	1,050 injured only	316	6	0.03
Recovery Regressor	CatBoost	MAE	1,050 injured only	150	6	0.03

Note: Tree counts are medians derived from 5-fold CV—not tuned on final fit.

Strict Leakage-Free 5-Fold Cross-Validation



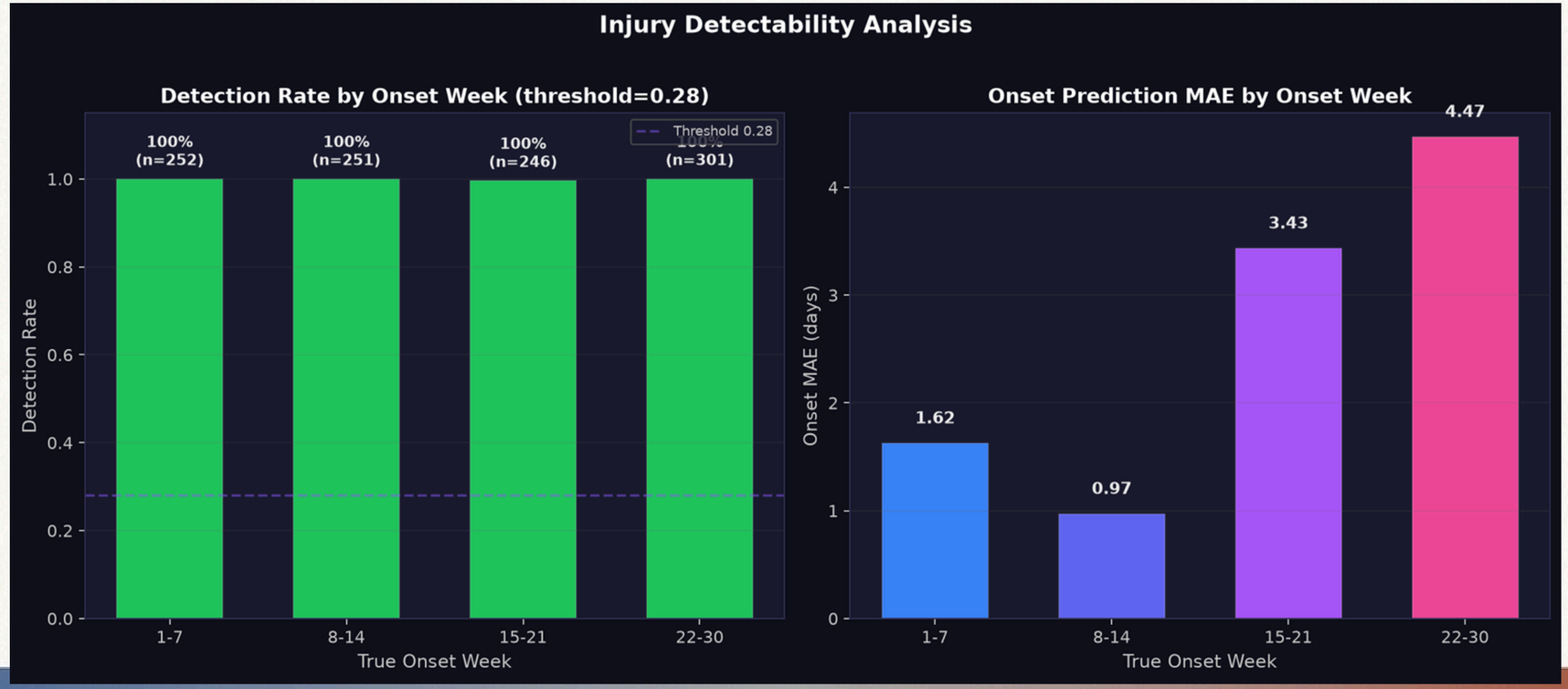
Each athlete gets exactly one prediction from a model that never saw them. Folds stratified on injury label (2,400 train / 600 val). Baselines computed strictly per-fold.

Fold	F1 @ 0.28	Onset MAE	Recovery MAE
1	0.5303	2.452	2.838
2	0.5243	2.400	3.029
3	0.5276	2.810	3.057
4	0.5237	2.957	2.857
5	0.5245	2.905	2.933
Mean	**0.5261**	**2.705**	**2.943**

Maximum F1 spread across folds is only 0.0066. These results are incredibly stable, not a statistical anomaly.

Key Insight: The Detection Horizon

Correlation between predicted probability and true onset day is an **extreme −0.852**.



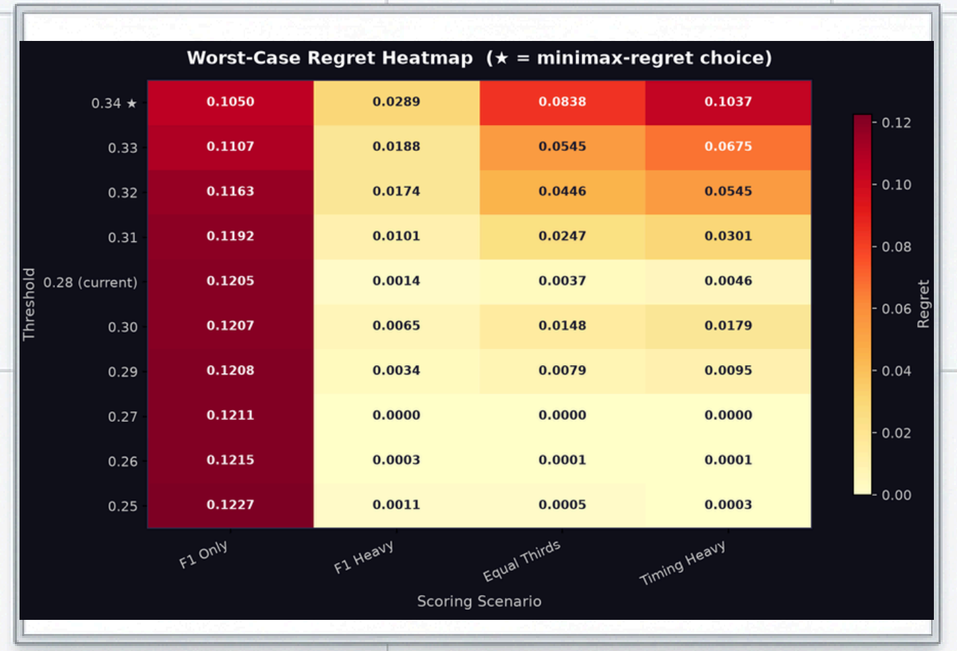
Week 1 (Days 1-7)

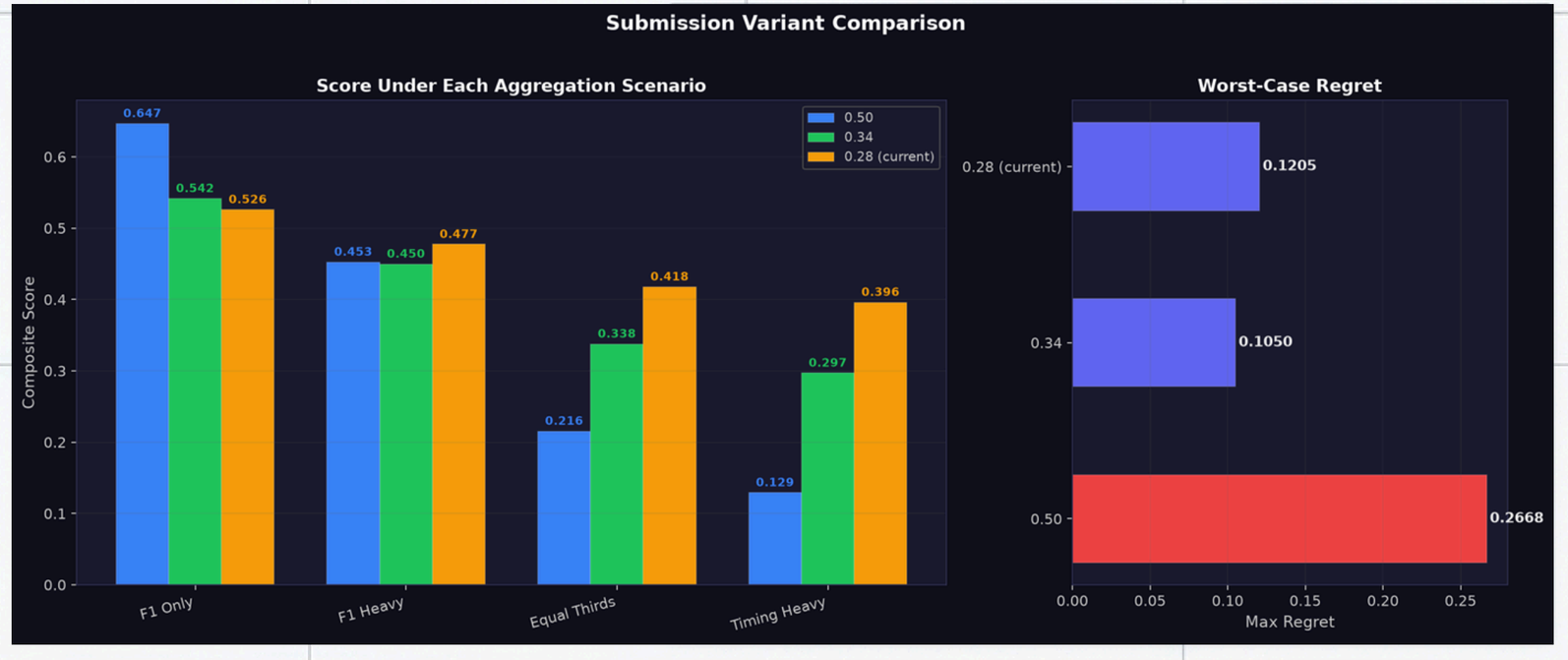
Week 4 (Days 22-30)

The model is a **genuine two-week early-warning system**. Late-onset athletes (days 22–30) are statistically **indistinguishable from healthy athletes at Day 30**. The physiological signal has a rigid **~14-day horizon**.

The Threshold Decision (Game Theory)

The Strategic Asymmetry: Missing an injury incurs a massive 30-day penalty.
Missing an injury is far worse than a false alarm.





Threshold Comparison:
0.50 (Standard): F1 = 0.6466 | Onset Skill = 0.0000 | Recall = 50.2%
0.34 (Minimax): F1 = 0.5416 | Onset Skill = 0.4717 | Recall = 94.9%
0.28 (Selected): F1 = 0.5261 | Onset Skill = 0.6416 | Recall = 99.9%

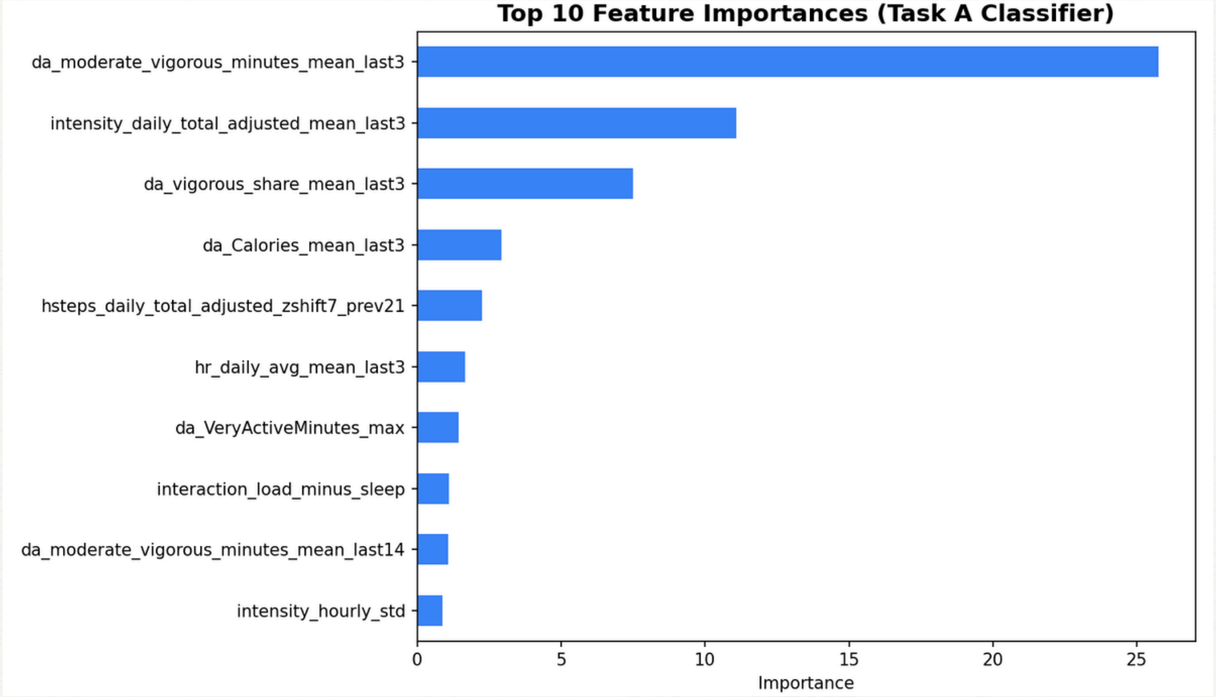
Synthesis: Moving the threshold from **0.50 to 0.28** sacrifices **0.12** in classification F1, but **unlocks a massive +0.73 gain across timing skills**. F1 would need to carry **>75%** of the total scoring weight for 0.50 to be viable.

Performance Results

0.5261	0.7655	0.6416	0.0857
F1 Score	ROC AUC	Onset Skill	Recovery Skill

	Predicted Healthy	Predicted Injured
Actually Healthy	TN = 61	FP = 1,889
Actually Injured	**FN = 1**	TP = 1,049

Precision: 0.3570 | Recall: 0.9990 | Pure Onset MAE: 2.705 days | Pure Recovery MAE: 2.943 days



The system flagged 97.9% of athletes for review, missing exactly **ONE injured** athlete out of 1,050.

What Sets Us Apart

1. Discovery, Not Just Prediction



Uncovered a fundamental physiological truth—injury detectability possesses a rigid ~14-day horizon. Late-onset unpredictability is biology, not a tuning failure.

2. Game-Theory Optimization



Formally analyzed the asymmetric 30-day timing penalty. Deriving the optimal 0.28 threshold via break-even analysis delivered +0.20 to the composite score—outperforming any model tuning.

3. Zero-Leakage Guarantee



Strict nested 5-fold CV with per-fold baselines and zero global imputation. A microscopic F1 spread of 0.0066 guarantees high out-of-sample reliability.

4. 347 Athlete-Relative Features



Features explicitly compare athletes against their own historical disjoint time windows, creating a genuinely personalized, rather than population-based, warning system.

FaultLine is not just a classifier - it’s a calibrated early-warning system with a known, validated detection range.

Negative Results & Learnings

Feature Ablation Limits

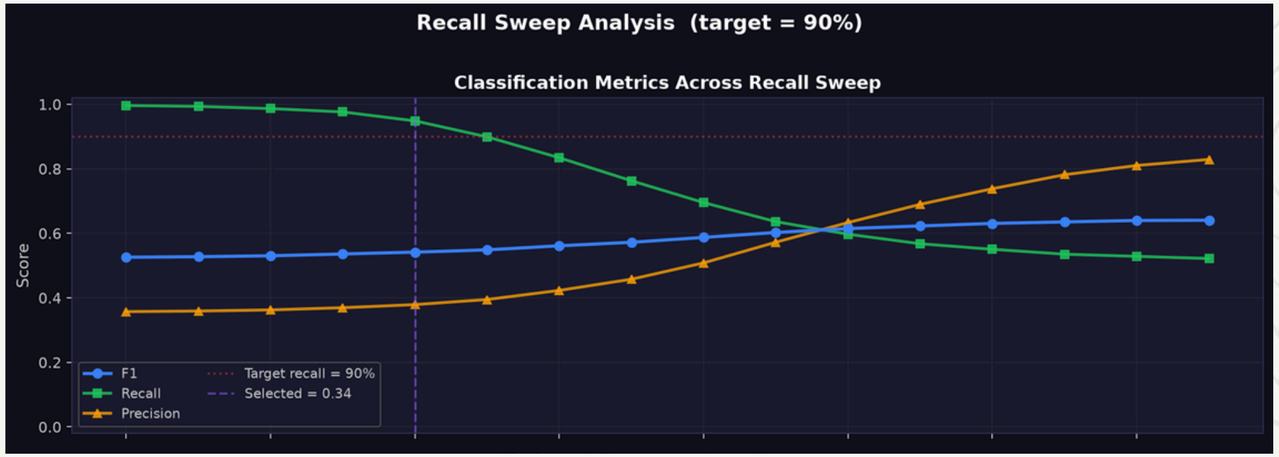
No single feature family generated repeatable standalone improvements; all 347 features contribute marginally to the final ensemble.

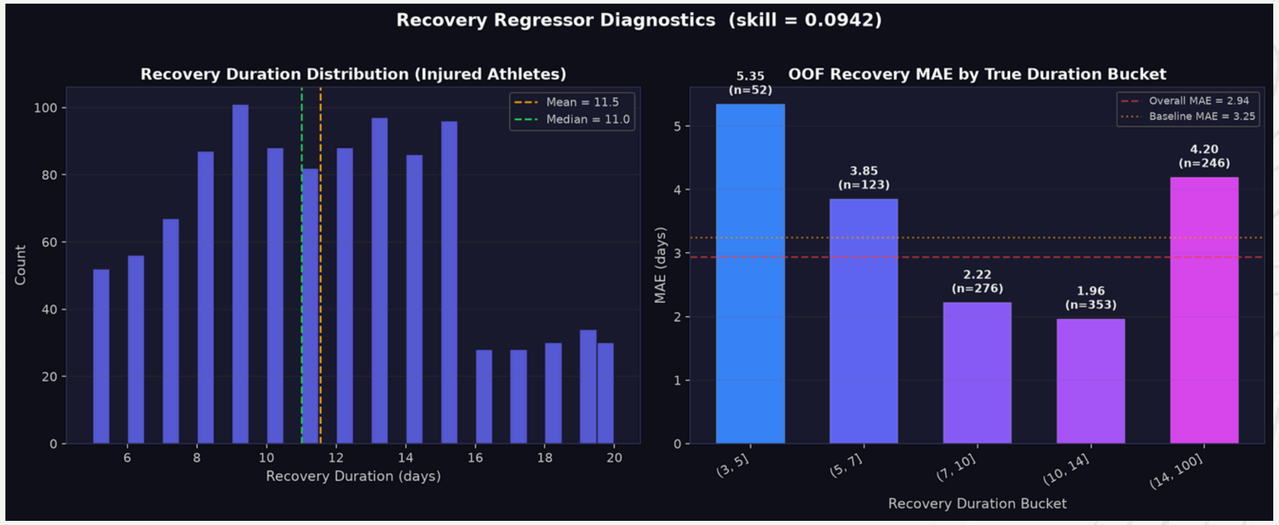
Nested Tuning Noise

Fully nested configuration selection ultimately scored worse (AUC 0.7505 vs 0.7655), revealing that micro-tuning was simply fitting noise over gaps of ~0.005.

Recovery Regressor Ceiling

8 configurations tested. Linear models performed poorly (ElasticNet 3.02 MAE, Ridge 3.04). Baseline MAE (3.249) sits virtually identical to the best constant predictor (3.233), capping max achievable skill at ~0.15.





Conclusion: Honesty builds trust. The pipeline is operating at the absolute ceiling of this observation window. Remaining modeling headroom is bounded at ≈ +0.03.

FaultLine is not just a classifier - it is a deeply calibrated early-warning system with a mathematically validated detection range and optimized risk threshold.

FaultLine

Team: Ctrl Freaks